Tutorial 3

$$\int \frac{1}{(x^{1/3}-1)x^{1/4}}\,dx$$

Ex. 8.5

Q46

$$\int \frac{1}{(x^{1/3}-1)\sqrt{x}}\,dx$$

~~Hint: $x = u^6$~~

$u = \sqrt{x} \Rightarrow du = \frac{1}{2\sqrt{x}}\,dx$

$\Downarrow$

$x = u^2$

$x^{1/3} = u^{2/3}$

$$= 2\int \frac{1}{u^{2/3}-1}\,du$$

$t = u^{1/3} \Rightarrow u^{2/3} = t^2$

$dt = \frac{1}{3}u^{-2/3}\,du$

$= \frac{1}{3}\left(\frac{u^{1/3}}{t}\right)^{-2} du$

$= \frac{1}{3}t^{-2}\,du$

$du = 3t^2\,dt$

$$= 2\int \frac{3t^2\,dt}{t^2-1}$$

$$= 6\int \frac{t^2}{t^2-1}\,dt$$

$$\frac{t^2}{t^2-1} = \frac{(t^2-1)+1}{t^2-1} = 1 + \frac{1}{t^2-1}$$

$$\frac{1}{t^2-1} = \frac{A}{t-1} + \frac{B}{t+1}$$

Q48

$$\int \frac{1}{x\sqrt{x+9}}\,dx$$

$$= \int \frac{2u\,du}{(u^2-9)\,u}$$

$$= \int \underbrace{\frac{2}{u^2-9}}_{\text{P.F.}}\,du$$

$u = \sqrt{\cdots}$

$u = \sqrt{x+9}$

$u^2 = x+9$

$x = u^2 - 9$

$dx = 2u\,du$

Q53

$$\int \frac{\overset{u}{\boxed{\sqrt{1+\sqrt{x}}}}}{\underset{(u^2-1)^2}{\boxed{x}}}\,\overset{4u(u^2-1)\,du}{\boxed{dx}}$$

"Rationalization"

$u = \sqrt{1+\sqrt{x}}$

$u^2 = 1+\sqrt{x}$

$\sqrt{x} = u^2 - 1$

$x = (u^2-1)^2 \Rightarrow dx = 2(u^2-1)\cdot 2u\,du$

$= 4u(u^2-1)\,du$

Q50 $\displaystyle\int \frac{1}{x^6(x^5+4)}\,dx \cdot \frac{x^4}{x^4}$

Q49

$\displaystyle\frac{1}{x(x^4+1)} \cdot \frac{x^3}{x^3}$

$$= \int \frac{x^4}{x^{10}(x^5+4)}\,dx$$

$u = x^5 \Rightarrow x^{10} = u^2$

$du = 5x^4\,dx$

$$= \frac{1}{5}\int \frac{1}{u^2(u+4)}\,du$$

$$\frac{1}{u^2(u+4)} = \frac{A}{u} + \frac{B}{u^2} + \frac{C}{u+4}$$

General: $\displaystyle\int \frac{dx}{x^m(x^n+a)}$

If $m+n-1$ is a multiple of n, try mult. by $\dfrac{x^{n-1}}{x^{n-1}}$.

Ex 8.8

Q42 $\displaystyle\int_1^2 \frac{1}{x\ln x}\,dx = \lim_{c\to 1^+}\int_c^2 \frac{1}{x\ln x}\,dx$

$(x=1)$

$$= \lim_{c\to 1^+}\Big[\ln|\ln|x||\Big]_c^2$$

$$= \lim_{c\to 1^+}\Big[\ln(\ln 2) - \ln|\ln c|\Big]$$

$$= \infty$$

$\displaystyle\int \frac{1}{x\ln x}\,dx$ $\quad u=\ln x$

$du = \frac{1}{x}dx$

$$= \int \frac{1}{u}\,du$$

$$= \ln u + C$$

$$= \ln(\ln x) + C$$

~~$c\to 0^+$~~

~~$\ln c \to -\infty$~~

~~$\ln|\ln c| \to \infty$~~

~~$-\ln|\ln c| \to -\infty$~~

~~$\therefore \int_1^2 \frac{1}{x\ln x}\,dx$ diverges.~~

$c\to 1^+$

$\ln c \to 0$

$\ln|\ln c| \to -\infty$

$-\ln|\ln c| \to \infty$

$\therefore \int_1^2 \frac{1}{x\ln x}\,dx$ diverges.

Q50

$$\int_{-1}^{1} -x\ln|x|\,dx = \int_{-1}^{0} -x\ln|x|\,dx + \int_{0}^{1} -x\ln|x|\,dx$$

$(x=0)$

$$= \int_{-1}^{0} -x\ln(-x)\,dx + \int_{0}^{1} -x\ln x\,dx$$

$u=-x$

$du=-dx$

$\boxed{x=0,\ u=0}$ → WRONG!!

$x=-1,\ u=1$

$$= -\int_{1}^{0} u\ln u\,du - \int_{0}^{1} x\ln x\,dx$$

$$= \int_{0}^{1} u\ln u\,du - \int_{0}^{1} x\ln x\,dx$$

$$= 0$$

$f(x)$ is odd

$\Rightarrow \int_{-a}^{a} f(x)\,dx = 0$

This does not work for improper integrals.

Counterexp.

$\int_{-1}^{1} \frac{1}{x}\,dx$.

$f(x)=\frac{1}{x}$ is odd, but $\int_{-1}^{1}\frac{1}{x}\,dx = \int_{0}^{1}\frac{1}{x}\,dx + \int_{-1}^{0}\frac{1}{x}\,dx$ diverges.

$$= \lim_{c \to 0^-} \int_{-1}^{c} -x \ln(-x)\, dx$$

$$+ \lim_{b \to 0^+} \int_{b}^{1} (-x \ln x)\, dx$$

$$\vdots$$

$$= -\frac{1}{4} + \frac{1}{4}$$

$$= 0$$

$c \to 1^-$

$1 - c \to 0$

$\ln|1-c| \to -\infty$

$-\ln|1-c| \to +\infty$

<u>Q48</u> $\displaystyle\int_0^2 \frac{dx}{1-x} = \int_0^1 \frac{dx}{1-x} + \int_1^2 \frac{dx}{1-x}$

$(x = 1)$

$$\int_0^1 \frac{dx}{1-x} = \lim_{c \to 1^-} \int_0^c \frac{dx}{1-x} = \lim_{c \to 1^-} \Big[-\ln|1-x|\Big]_0^c$$

$$= \lim_{c \to 1^-} \Big[-\ln|1-c|\Big] = \infty$$

Since $\int_0^1 \frac{dx}{1-x}$ div., $\int_0^2 \frac{dx}{1-x}$ div.

Q59

$$\int_\pi^\infty \frac{2+\cos x}{x}\,dx$$

$-1 \leq \cos x \leq 1$ for $x \geq \pi$

$1 \leq \cos x + 2 \leq 3$

$$\boxed{\frac{1}{x} \leq \frac{\cos x + 2}{x}} \leq \frac{3}{x} \quad \because x \geq \pi > 0$$

$\boxed{\int_\pi^\infty \frac{1}{x}\,dx \text{ diverges}}$

Since $\int_\pi^\infty \frac{1}{x}\,dx$ div, $\int_\pi^\infty \frac{2+\cos x}{x}\,dx$ div. by Direct Comparison Test.

$$\int_\pi^\infty \frac{2+\cos x}{x}\,dx = \underbrace{\int_\pi^\infty \frac{2}{x}\,dx}_{\text{div.}} + \int_\pi^\infty \frac{\cos x}{x}\,dx$$

$\therefore \int_\pi^\infty \frac{2+\cos x}{x}\,dx$ div. because $\int_\pi^\infty \frac{2}{x}\,dx$ div.

Not True

counterexp.

$$\int_a^\infty \frac{1}{x} - \frac{1}{x+1}\, dx$$

$$= \int_a^\infty \frac{1}{x}\, dx - \int_a^\infty \frac{1}{x+1}\, dx$$

$$\frac{1}{x} - \frac{1}{x+1}$$

$$= \frac{x+1-x}{x(x+1)}$$

$$= \frac{1}{x(x+1)} \sim \frac{1}{x^2}$$